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Technical Note

Adaptive Kalman Filter-Based Impulsive Noise Cancellation for Broadband Active Noise Control in Sensitive Environments

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Abstract

Impulsive noise poses a significant challenge to broadband feedforward active noise control (ANC) systems, particularly in sensitive environments such as infant incubators. This paper presents an adaptive impulsive noise cancellation approach based on the Kalman filter, designed to improve noise attenuation performance under nonstationary and impulsive interference. The proposed framework integrates impulsive noise detection with a Kalman filter-based suppression scheme. Simulation studies are conducted to evaluate the performance of the combined system in comparison to traditional ANC methods, such as Filtered-x Least Mean Square (FxLMS) and Filtered-x Normalized LMS (FxNLMS). Results demonstrate that the Kalman filter can effectively reduce the influence of impulsive disturbances without degrading overall broadband noise cancellation. A case study involving an infant incubator illustrates the practical effectiveness and robustness of the proposed technique in a real-world healthcare application. The findings support the integration of Kalman filter-based adaptive control in future ANC designs targeting impulsive noise environments.

Keywords: impulsive noise; active noise control; Kalman filter; least mean square (LMS); infant incubator

1. Introduction

Impulsive noise presents a significant challenge in active noise control (ANC) systems due to its transient, high-amplitude nature and unpredictable occurrence. Unlike stationary background noise, impulsive noise can disrupt the stability and convergence of adaptive algorithms commonly used in ANC, such as the Filtered-x Least Mean Squares (FxLMS) and its normalized variant, FxNLMS. These conventional algorithms based on the Wiener filter are primarily designed for stationary broadband noise and often fail to respond effectively to abrupt disturbances. This difficulty in handling complex, non-stationary signals is a common challenge across various domains, from medical environments to geophysical exploration [1–4].

Robust suppression of impulsive noise is particularly critical in acoustically sensitive environments. One such example is the neonatal intensive care unit (NICU), where incubators are used to protect premature, low-birth-weight, or severely ill newborns. The average noise level inside a NICU is 75 dB, which is much higher than the recommended noise level by the WHO (45 dB in the daytime and 35 dB in the nighttime); our proposed methods can effectively reduce the noise level inside incubators [5,6]. However, impulsive noise signals, for example, caused by closing a door or dropping medical tools on the floor, are associated

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with noises with peak levels of around 120 dB. Excessive exposure to sudden, sharp acoustic events in these settings has been associated with adverse developmental outcomes [7–11]. Moreover, due to the much higher amplitude of impulsive noise, it might cause instability to traditional FXNLMS algorithms and make the system diverge. Therefore, ensuring a low-noise environment is a priority, and traditional ANC systems may fall short of this requirement under impulsive noise conditions.

To address these limitations, researchers have explored various techniques for impulsive noise detection and mitigation, including fractional lower order power-based algorithms, threshold methods, and logarithmic and trigonometric transformation methods [12–14]. While these methods have shown effectiveness in controlled or electrically generated impulsive noise environments, they often assume an idealized impulse model—typically a sharp, delta-function–like spike similar to impulsive artifacts in electrical signals. However, acoustic impulsive noise in real clinical settings, such as the NICU, is far more complex. These sounds exhibit broader temporal spread, variable amplitude envelopes, and frequency-dependent decay characteristics that differ fundamentally from the instantaneous impulses assumed in many existing algorithms. Moreover, real-world acoustic impulsive noise frequently coexists with broadband ambient noise (e.g., ventilation systems, alarms, and caregiver activity), creating a mixed and dynamic noise environment.

In recent years, Kalman filter-based approaches have gained increasing attention in active noise control (ANC) due to their strong capability in tracking time-varying systems and handling dynamic acoustic environments [15–20]. Liebich et al. demonstrated the effectiveness of time-domain Kalman filtering for real-time ANC in headphone applications [16]. Lopes et al. further extended the Kalman filter to multichannel ANC using a random-walk model to improve tracking performance [20]. To reduce computational burden, Fabry et al. proposed a reduced-complexity Kalman filter for practical ANC systems [19]. More recently, Luo et al. introduced a hybrid CNN–Kalman framework for generative fixed-filter ANC, showing improved performance in nonstationary conditions [15]. Aboutiman et al. applied Kalman-based ANC to encapsulated structures with non-minimum phase characteristics, validating its robustness in challenging acoustic paths [17]. These studies collectively highlight the growing role of Kalman filtering in modern ANC, particularly for adaptive control in complex and dynamic noise environments [18].

In this paper, we propose an adaptive impulsive noise cancellation framework that integrates a real-time detection module with a Kalman filter-based control scheme. Kalman filtering has shown promise due to its recursive nature and ability to model dynamic systems under uncertainty. When integrated with a detection mechanism, the Kalman filter can selectively suppress impulsive noise without significantly degrading the performance of the ANC system under normal operating conditions. This approach enhances the robustness of broadband feedforward ANC systems under nonstationary and impulsive acoustic environments. We compare the performance of our method with conventional FxLMS and FxNLMS algorithms through simulations.

To demonstrate the practical applicability of our approach, we apply the proposed framework to a simulated NICU incubator environment. Experimental results confirm that the Kalman filter-based method significantly improves impulsive noise suppression while maintaining overall ANC performance. This case study highlights the potential of the technique for broader use in medical, industrial, and consumer applications where impulsive noise is a concern.

The main contribution of this study aims to bridge a critical gap in clinical noise mitigation by initially developing a realistic impulsive noise detection framework grounded in actual NICU acoustic characteristics. To address the dynamic, mixed-noise challenge of the NICU in real time, we propose an adaptive, Kalman-filter-based noise control algorithm.

Our approach features an integrated detection-and-control method that is designed to robustly manage real-world acoustic impulses, moving beyond reliance on highly idealized models. Ultimately, a clinically grounded evaluation suggests this methodology offers improved noise suppression capabilities for various healthcare applications.

The remainder of this paper is organized as follows. Section 2 presents the impulsive noise detection technique. Section 3 introduces the general ANC framework and proposes a Kalman filter-based feedforward ANC scheme. Section 4 describes the case study, while Section 5 discusses the simulation results. Finally, Section 6 concludes the paper.

2. Impulsive Noise Detection

The proposed impulsive noise cancellation framework consists of three main components: (1) impulsive noise detection, (2) Kalman filter-based adaptive noise control, and (3) integration with conventional ANC algorithms for hybrid operation. The goal is to enhance ANC system performance in environments where impulsive noise is present, using the NICU incubator as a representative application.

To enable adaptive response to transient disturbances, impulsive noise must first be reliably detected. The detection algorithm monitors the error signal or input reference for abnormal amplitude spikes using time-domain statistics. A thresholding method is employed, where the signal is compared against a dynamic threshold based on short-term energy or amplitude deviation. When a spike is detected above the threshold, it is classified as impulsive noise.

2.1. Impulsive Noise Detection Using Normalized Power Sequence

Impulsive spikes have significantly larger amplitude values compared to normal signals. Therefore, the signal power of certain time lengths can be used to detect impulsive noise [21]. First, a digitalized signal can be divided into several frames, and each frame contains a few samples of certain time periods. Hence the power for kth frame is calculated as follows:

$$p(k) = \sum_{i=0}^{N-1} x^2(\mathrm{i} + kN) \quad (1)$$

Then, we calculate the normalized power by blocks, with the length of each block being L:

$$p_n(j,k) = \frac{p_w(j,k) - \min(p_w(j,k))}{\max[p_w(j,k) - \min(p_w(j,k))]} \quad (2)$$

where $p_w(j,k)$ is the window's power.

Finally, the standard deviation of the normalized power sequence in one block is

$$p_v(j,k) = \frac{\sum_{j=0}^{L-2}[p_n(j,k) - \mathrm{mean}(p_n(j,k))]}{L-1} \quad (3)$$

The square root of the variance $\sqrt{p_v(k)}$ will decrease dramatically when the impulsive noise ends.

Therefore, $\begin{cases} \sqrt{p_v(k)} < \text{threshold} & \text{normal background noise} \\ \sqrt{p_v(k)} \geq \text{threshold} & \text{impulse noise} \end{cases}$.

2.2. Impulsive Noise Detection Using Conditional Median Filter

Median filters are widely used for removing impulsive noise contained in image and audio files. The output of a median filter is the median value of the ordered input sequences:

$$mf(k) = median\{x(i)\},\ i = k - L + 1\ to\ Kl \quad (4)$$

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Here L is the number of input sequences. From the above formula, it can be found that the output of the median filter has a delay. The delay equals $\frac{L-1}{2}$. This indicates that the median filter will remove the impulsive spikes if the impulsive noise duration is less than $\frac{L-1}{2}$ [21].

Certain thresholds can be applied to remove impulsive noise if the statistical information of the noise is known. This will produce a modified median filter, which is called a conditional median filter [22]. The detail is shown in the following formula:

$$cmf(k) = \begin{cases} mf(k),\ if\ |mf(k) - x(k-d)| > threshold \\ x(k-d),\ otherwise \end{cases} \tag{5}$$

with $d = \frac{L-1}{2}$. In the simulations, the mean value in one window block can be chosen as the threshold of the conditional median filter.

2.3. Impulsive Noise Detection by Signal-Dependent Rank-Order Mean Algorithm

The main disadvantage of the median filter and conditional median filter is that the sample number of input sequences increases when the signal is corrupted by heavy impulsive noise with a long duration. This will dramatically increase the algorithm computational cost.

The signal-dependent rank-order mean algorithm (SD-ROM) was originally proposed to remove the impulsive noise from the image files. Here it has been modified to a one-dimensional version (Algorithm 1). The algorithm procedures are as follows [23]:

Algorithm 1: SD-ROM Algorithm

1. Suppose one five-sample window is sliding from the beginning of input signal to the end. For example, the sequence in one window is x1, x2, x3, x4, and x5.
2. One sequence is constructed to save the input samples as follows: $y_i = x_i$. Then it will be sorted with increasing order. The result is $r_1\ r_2\ r_3\ r_4\ r_5$
3. Then, the rank-ordered differences d_i are calculated by

$$d_i = \left\{ \begin{array}{ll} r_i - r_3 & if\ r_3 < \mu \\ x_3 - r_{4-i} & if\ r_3 > \mu \end{array} \right\}$$

with $\mu = \frac{r_2+r_3}{2}$.

4. Finally, the algorithm detects x_3 s, an impulsive sample, if any of the following two conditions are satisfied.

$$d_i > T_i,\ i = 1,2$$

where the threshold T_i is the experimental values for the simulation.

3. Kalman Filter-Based Impulsive Noise Control

3.1. Brief Introduction of Feedforward ANC

A conventional feedforward ANC system is shown in Figure 1. It can effectively to reduce the wideband stationary ambient noise by update the adaptive filter by using the LMS algorithm [23,24].

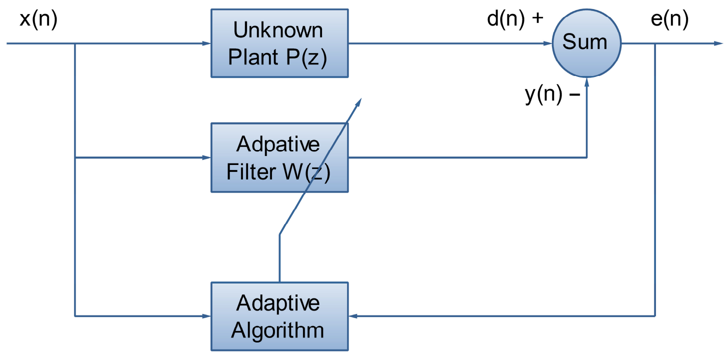

Figure 1. Block diagram of feedforward ANC system. The reference sensor captures the input signal, which is processed by an adaptive filter. The filter's output is combined with the primary noise, and the error sensor measures the residual noise. This error signal is then fed back to update the adaptive filter.

The reference sensor picks up the is the reference noise signal x(n), P(z) is an unknown plant, d(n) is the primary noise, and the adaptive filter W(z) generates anti-noise y(n). The error in this system is

$$e(n) = d(n) - y(n) \tag{6}$$

The primary noise d(n) can be modeled as the reference noise after passing through a FIR filter, which is present in the primary path.

$$d(n) = x(n) \ * \ p(n) = \sum_{m=0}^{M-1} p_m(n)x(n-m)$$

where $p(n)$ is the impulse response of the primary path $P(z)$, M is the length of the impulse response, and * denotes linear convolution. $P(z)$ is the Fourier transform of p(n).

$$P(z) = \sum_{m=0}^{M-1} p_m z^{-i}$$

and the corresponding impulse response $p(n)$ can be written as a vector, $\boldsymbol{p}(n) = [p_0(n) \ p_1(n) \ldots p_{M-1}(n)]^T$.

The error signal at the error sensor can be presented as

$$e(n) = d(n) - s(n) \ * \ y(n) = d(n) - \sum_{q=0}^{Q-1} s(q)y(n-q) \tag{7}$$

where $s(n)$ is the impulse response of secondary path $S(z)$ at time n and $*$ denotes linear convolution. The adaptive filter output is computed as

$$y(n) = \sum_{l=0}^{L-1} w(l)x(n-l) = w^T(n)x(n) \tag{8}$$

where $\boldsymbol{w}(n) = [w_0(n)w_1(n) \ \ldots \ w_{L-1}(n)]^T$ is the coefficient vector of the adaptive filter $W(z)$ of length L, and $\boldsymbol{x}(n) = [x(n) \ x(n-1) \ \ldots \ x(n-L+1)]^T$ is the reference signal vector.

The adaptive weight vector is updated by the FXLMS algorithm, expressed as [24]

$$w \ (n+1) = w \ (\mathrm{n}) + \mu \mathrm{x}'(n)e(n) \tag{9}$$

where is the step size and $x'(n) = [x'(n)\ x'(n-1) \ldots x'(n-L+1)]^T$ is the filtered reference signal vector by the estimated secondary path [3]. The filtered signal for $x'(n)$ is computed as

$$x'(n) = \hat{s}(n) * x(n) = \sum_{l=0}^{L-1} \hat{s}(l)x(n-l), \tag{10}$$

where $\hat{s}(n)$ is the impulse response of the secondary-path estimation filter, $\hat{S}(z)$.

For impulsive noise, we propose using the Kalman filter instead of the LMS algorithm to suppress the noise signal, as shown in Figure 2.

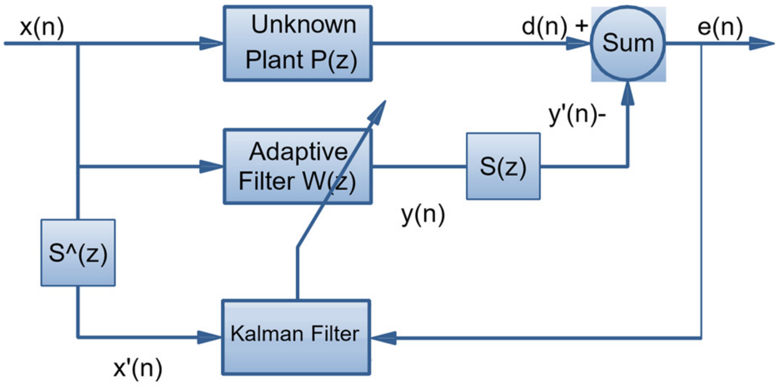

Figure 2. Impulsive noise mitigation using the Kalman filter algorithm. The adaptive filter is updated through the Kalman filter framework, while the filtered-X algorithm is applied to compensate for the secondary-path effect.

3.2. Brief Review of Kalman Filter

The Kalman filter is a recursive optimal estimator designed for linear dynamic systems under Gaussian noise [25–28]. It estimates the internal state of a system from noisy measurements by predicting the system's next state and updating the prediction using incoming observations.

The process consists of two main steps: prediction and updates [25].

In the first step, the state prediction is

$$\hat{w}(k|k-1) = \mathbf{A}\hat{w}(k|k-1) + \boldsymbol{B}\mu(k-1) \tag{11}$$

where $\hat{w}(k|k-1)$ is the predicted state estimate at time k, $\mathbf{A}$ is the state transition matrix, $\mathbf{B}$ is the control matrix, and $\mu(k-1)$ is the control input.

And the covariance prediction is [25,26]

$$\mathbf{P}(k|k-1) = \mathbf{A}\mathbf{P}(k|k-1)\mathbf{A}^{\boldsymbol{T}} + \mathbf{Q} \tag{12}$$

where $\mathbf{P}(k|k-1)$ is the predicted error covariance, and $\mathbf{Q}$ is the process noise covariance.

The next step is measurement update. The Kalman gain can be obtained by

$$\mathbf{K}(k) = \frac{\mathbf{AP}(\mathrm{k}|k-1)\mathbf{H}^T}{\mathbf{HP}(\mathrm{k}|k-1)\mathbf{H}^T + \mathbf{R}} \tag{13}$$

with $\mathbf{H}$ is the observation matrix and $\mathbf{R}$ is the measurement noise covariance [28].

Then the state can be updated as

$$\hat{w}(k|k) = \hat{w}(k|k-1) + \mathbf{K}(k)(z_k - \mathbf{H}\hat{w}(k|k-1)) \tag{14}$$

where $\hat{w}(k|k)$ is updated state estimate and z_k is the measurement at time k.

And the covariance matrix can be updated as follows:

$$\mathbf{P}(k|k) = (\mathbf{I} - \mathbf{K}(k))\mathbf{P}(k|k-1) \tag{15}$$

3.3. Combine Kalman Filter with ANC

In the section, we combine the Kalman filter with an ANC system to address the issue of impulsive noise. After impulsive noise detection, if the reference microphone senses traditional ambient noise, the FxLMS algorithm will be triggered, and the Kalman filter state will be updated as well. Once impulsive noise is detected, the Kalman filter will be used to cancel the impulsive noise. The complete combined scheme is detailed in Algorithm 2.

Algorithm 2: Impulsive Noise Detection and Control

Input: reference signal noise x[n]
Output: anti-noise y[n]

1. If x[n] is impulsive noise
2.   Use Kalman filter to generate anti-noise y[n]
3. Else if x[n] is not impulsive noise
4.   Use NLMS filter to generate y[n]
5. y[n] go through the secondary pass
6.   Error sensor picks up residue noise e[n]
7. Use e[n] to update NLMS filter's coefficient
8. Use e[n] to update Kalman filter's state
9. Go to 1

When the impulsive noise detection result is positive, we apply the Kalman filter algorithm to update and filter the input non-stationary noise. The state of the system is the adaptive filter, w(n), and residual disturbance, e(n), which is the error signal and can be written as

$$\mathrm{w}(n+1) = \mathbf{A}\mathbf{w}(\mathrm{n}) + \boldsymbol{B}r(n) \tag{16}$$

$$e(n) = \boldsymbol{K}\boldsymbol{w}(n) + \boldsymbol{P}\boldsymbol{r}(n) \tag{17}$$

where

$$\mathbf{A} = \alpha \begin{bmatrix} 1 & \dots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \dots & 1 \end{bmatrix}_{M\times M} \quad \textit{and } \mathbf{B} = \begin{bmatrix} 1 & \dots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \dots & 1 \end{bmatrix}_{M\times M}$$

The Kalman filter based feedforward ANC system is explained by the mapping from the ANC problem to the Kalman filter state-space model:

The state vector **w**(n) is the ANC filter coefficient vector. The measurement e(n) is the error microphone signal. The observation matrix **C**(n) is $\mathbf{x}'(\mathrm{n})^{\mathrm{T}}$ (the transposed filtered reference vector). We assume the process noise covariance **Q** = **I** and the measurement noise covariance **R** = **I**, corresponding to uncorrelated (diagonally dominant) processes and measurement noises with unit variance. This is a standard simplifying assumption when no specific correlation structure or variance information is available. The detailed operational flow of the Kalman filter–based ANC system is described in Algorithm 3.

Algorithm 3: Kalman Filter-Based ANC System

Input: reference signal noise x[n]
Output: updated filter coefficient w[n+1] the state function

1. Initialize the system with $\hat{x}(0|y_0) = \gamma$ and $\mathbf{K}(0) = \boldsymbol{\varphi}$ where γ and $\boldsymbol{\varphi}$ are small numbers but never zero.
2. Calculate the measured error signal $e(n)$, $e(n) = d(n) - y'(n)$
3. The Kalman gain is $\mathbf{G}(n) = \frac{\mathbf{A}\mathbf{K}(n,\, n-1)\mathbf{C}^T(n)}{\mathbf{C}(n)\mathbf{K}(n,\, n-1)\mathbf{C}^T(n)+\mathbf{R}(n)}$ (18)
4. the innovation vector is $\boldsymbol{\alpha}(n) = e(n) - \mathbf{C}(n)\hat{\mathbf{w}}(n|e_{n-1})$ (19)
5. Update the coefficients of the adaptive filter
$$\hat{\mathbf{w}}(n+1|y_n) = \mathbf{A}\hat{\mathbf{w}}(n|e_{n-1}) + \mathbf{G}(n)\boldsymbol{\alpha}(n) \quad (20)$$
6. Calculate the correlation matrix
$$\mathbf{K}(n) = \mathbf{K}(n, n-1) - \mathbf{A}\mathbf{G}(n)\mathbf{C}(n)\mathbf{K}(n, n-1) \quad (21)$$
7. Calculate the updated correlation matrix $\mathbf{K}(n+1, n) = \mathbf{A}\mathbf{K}(n)\mathbf{A}^T$
8. Go to 2

3.4. Integrated Hybrid Control Architecture

While Sections 2 and 3 detail the component algorithms for impulsive noise detection and control, respectively, the overall system relies on a tightly coupled, sequential hybrid architecture to manage the dynamic NICU environment. The detection block (Section 2) continuously processes the acoustic input, and its output directly gates and parameterizes the real-time processing of the control block (Section 3).

The flowchart in Figure 3, which illustrates the data and control flow, details how the input acoustic signal is first assessed for impulse characteristics before the adaptive Kalman filter determines the output noise reduction strategy. This integrated approach ensures the system manages mixed-noise conditions while specifically mitigating transient, high-energy impulses.

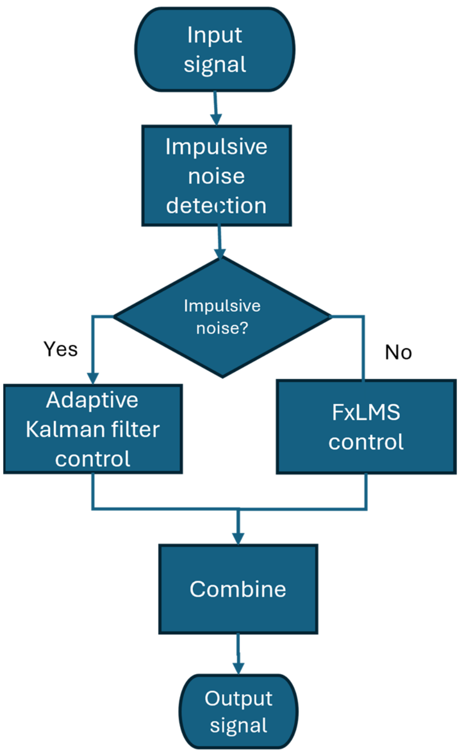

Figure 3. Workflow diagram illustrating the sequential process of the Integrated Hybrid Architecture, detailing the flow from impulsive noise detection to adaptive noise control.

4. Case Study

In this section, we present the case study based on infant incubator application. The neonatal intensive care unit is a special intensive care unit which takes care of very ill, very low-birth-weight neonates and preemies. An incubator is an apparatus used to maintain environmental conditions and mimic the womb so that infants obtain optimal conditions until their organs completely develop. Figure 4 shows the Giraffe infant incubator which is used as the test platform for this case study.

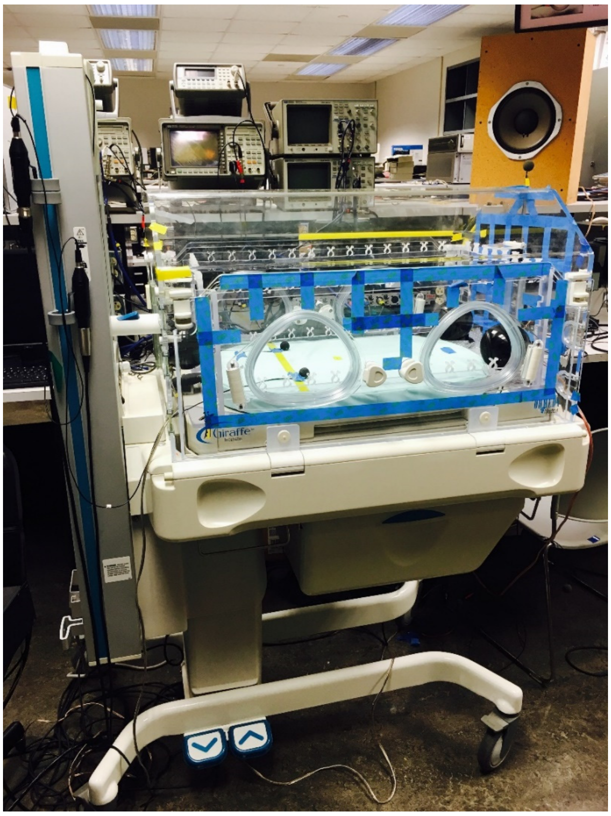

Figure 4. The Giraffe infant incubator. The diagram illustrates the placement of critical acoustic components: error sensors are positioned on the mattress near the infant's head; anti-noise speakers are located by the newborn's feet; a reference microphone is situated at the top of the incubator outside the enclosure; and an external speaker placed on the bench serves as the primary noise source.

While NICUs increase the survival rate of ill preterm infants dramatically, the advanced technical devices used in NICUs and the activities of the caregivers generate louder and more complicated noise. Research has revealed that exposure of infants to excessive noise significantly changes the physiological and behavioral responses of the infants. Researchers have tried to design active noise control (ANC) systems for incubators to reduce high-level noise [29–32]. However, applying ANC techniques to infant incubators in the NICU is much more difficult than usual because the noise field inside the incubator is much more complex than the noise field of typical ANC applications—it has multiple unexpected impulse noise sources with much higher noise levels compared to background noise. Unfortunately, impulse noises are non-stationary random signals and traditional ANC systems cannot cancel them effectively. Moreover, impulse noises can degrade the performance of a typical ANC system dramatically.

In this case study, we proposed using a conventional FxLMS algorithm to cancel background noise and using a Kalman filter to mitigate impulse noise. The noise used in this case study was recorded in a NICU in a hospital and consisted of sounds from alarms, people talking, doors opening and closing, writing on the top of incubator, and pages turning. The original sampling rate was 16 kHz and the length of it was 2 min and 5 s.

The time-domain characteristics of the acoustic environment within the NICU are critical for algorithm design. As shown in Figure 5, the recorded NICU noise consisted of a continuous background ambient noise floor interspersed with distinct impulsive events. These transient events are characterized by a significantly shorter time duration but a markedly higher amplitude—measured to be approximately five times greater than the steady-state background level—demonstrating the challenge of effectively managing mixed-noise conditions in this sensitive clinical setting.

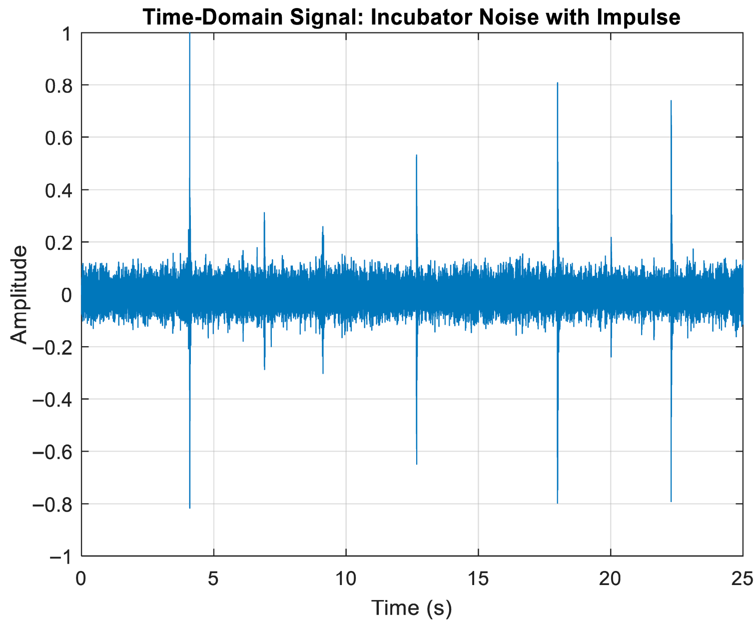

Figure 5. Time-domain waveform of NICU noise. The plot highlights impulsive noise occurrences at approximately 4, 13, 18, and 22 s. Compared to continuous background ambient noise, these impulsive events exhibit a markedly higher amplitude (approximately 5 times greater) and a significantly shorter time duration.

The spectral characteristics of the NICU's noise, detailed in Figure 6, directly informed the design of our hybrid system. The presence of a prominent, continuous background noise component, clearly visible across the time domain and centered around the 200–300 Hz range, dictates the need for frequency tracking capability in the core noise control algorithm. Mechanistically, this background noise drives the selection of the Kalman filter's state-space model and covariance parameters **Q**. Conversely, impulsive events are characterized by high energy spreading across a wideband frequency range. As these non-stationary characteristics cannot be accurately modeled or tracked by the linear state estimator, their existence necessitates a dedicated detection mechanism to identify and adaptively gate the filter's operation, thereby preserving system stability during transient acoustic events.

The testbed is shown in Figure 7; the reference microphone was placed on the top of the incubator. A loudspeaker was used as inside noise source and placed on the head side of the incubator. Two microphones were placed on the upper half of the incubator where they should have been around the infant's ears. Two loudspeakers were placed in two corners around the infant's feet towards two error microphones for safety reasons. In case any accidents happened to the loudspeakers, they were placed far away from the infant's ears.

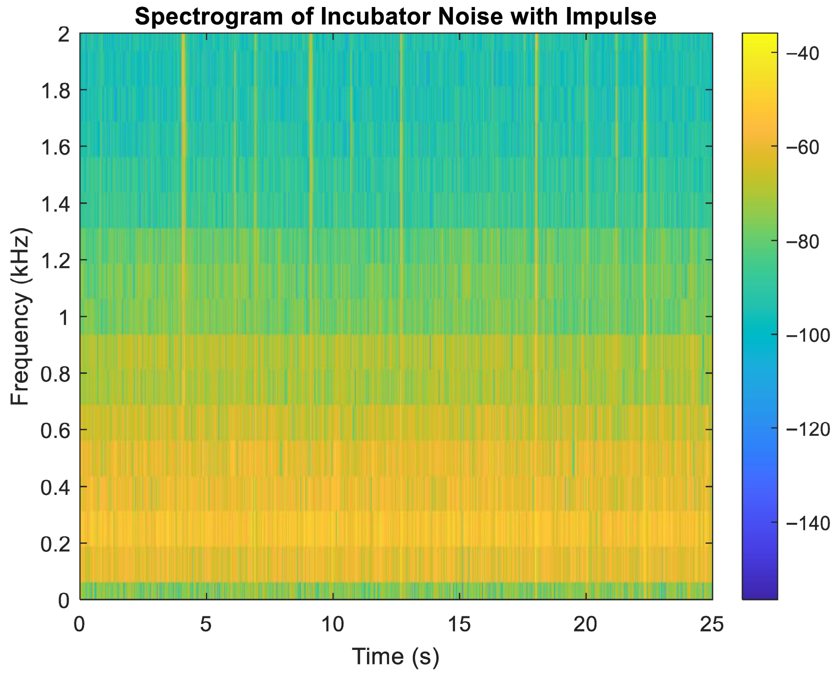

Figure 6. Spectrogram of the NICU acoustic environment. Corresponding to the impulsive noise events noted in Figure 5, high-amplitude energy (indicated by the yellow color) demonstrates wideband, short-duration characteristics, spreading across the system's entire frequency range (50 to 2000 Hz).

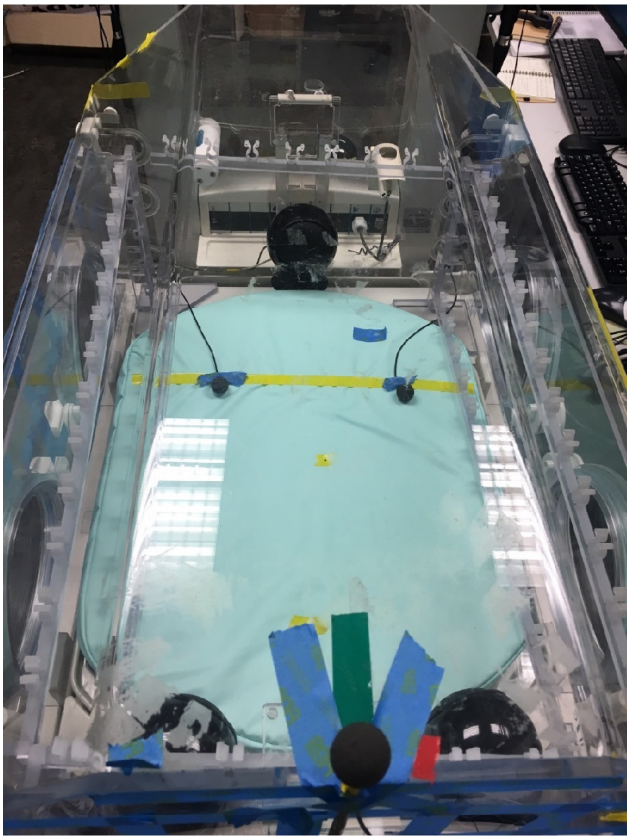

Figure 7. Test platform infant incubator with reference microphone, anti-noise speakers, error microphones.

All relevant transfer functions were measured directly from the physical setup and subsequently incorporated into a MATLAB R2014b model. This approach allowed the simulation to closely mimic the real system's behavior. Therefore, the results presented in this work can be regarded as simulation results based on experimentally derived parameters, providing a model-based representation of the experimental system.

In the experimental testbed, the setup was established within a real infant incubator to replicate a sensitive acoustic environment. The noise source was obtained from a recorded signal containing impulsive noise components, representative of realistic disturbances in clinical conditions. To reduce computational complexity during processing, the signal was downsampled to a sampling frequency of 3.2 kHz. The active noise control (ANC) configuration followed a $1 \times 2 \times 2$ structure, comprising one reference microphone, two control speakers, and two error microphones. The primary acoustic path, P(n), was modeled as a 128-tap discrete system, while both the secondary path and the Kalman filter were implemented with 64 taps to maintain causality and ensure stable adaptive control performance.

5. Results

This section presents the simulation results obtained from the experimental testbed, which was designed to replicate the acoustic characteristics of an infant incubator. The testbed integrates real sensors, actuators, and measured acoustic transfer functions to closely model the actual operating environment. Using these parameters, simulations were performed to evaluate the performance of the proposed hybrid adaptive ANC system under both impulsive and continuous noise conditions. The results provide a detailed assessment of the system's convergence behavior, noise attenuation capability, and overall stability.

5.1. Impulsive Noise Deteaction

The simulation was conducted using this processed signal as the input to evaluate the performance of the proposed algorithm. The corresponding results are presented in Figures 8 and 9.

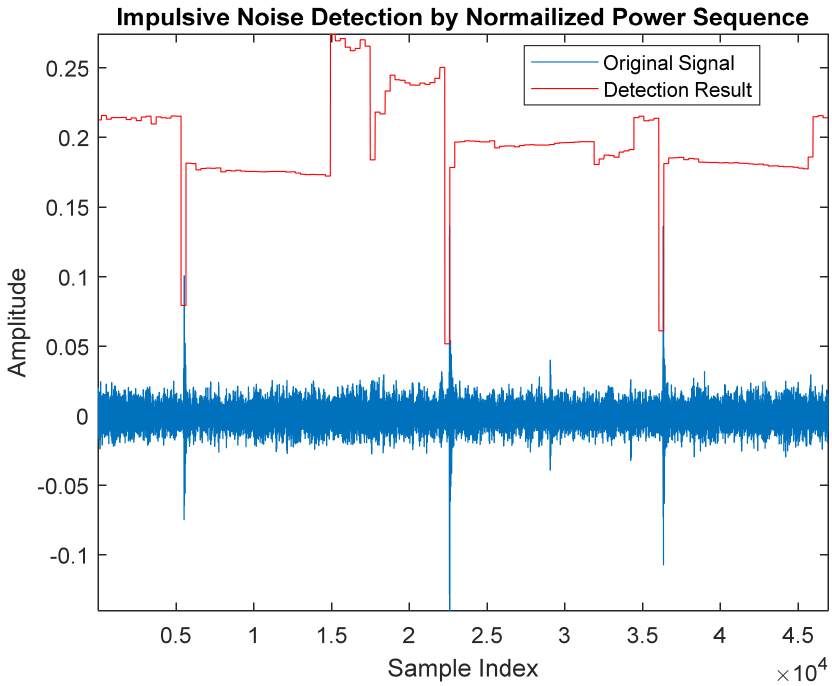

Figure 8. Impulsive noise detection performance using the normalized power sequence method when the noise source contains impulsive characteristics. The plot demonstrates that the dramatic, corresponding changes in the detection output (red line) accurately indicate the occurrence of the impulse noise signal (blue curve), confirming the high detection effectiveness of the algorithm.

The effectiveness and stability of the impulsive noise detection algorithm were verified under both impulse-present and steady-state conditions. As illustrated in Figure 8, when the noise source contains characteristic impulsive content, the detection output (red line) exhibits dramatic, high-amplitude excursions that align accurately with the peaks of the impulse noise signal (blue curve). This correspondence confirms the algorithm's high detection effectiveness. Conversely, Figure 9 displays the detection output when the acoustic environment is intentionally held free of impulsive noise. In this scenario, the detection output shows minimal fluctuations, successfully maintaining a near-zero baseline. This stability is crucial, as it confirms the algorithm's robustness and resistance to generating false positive detections under typical steady-state or background noise conditions.

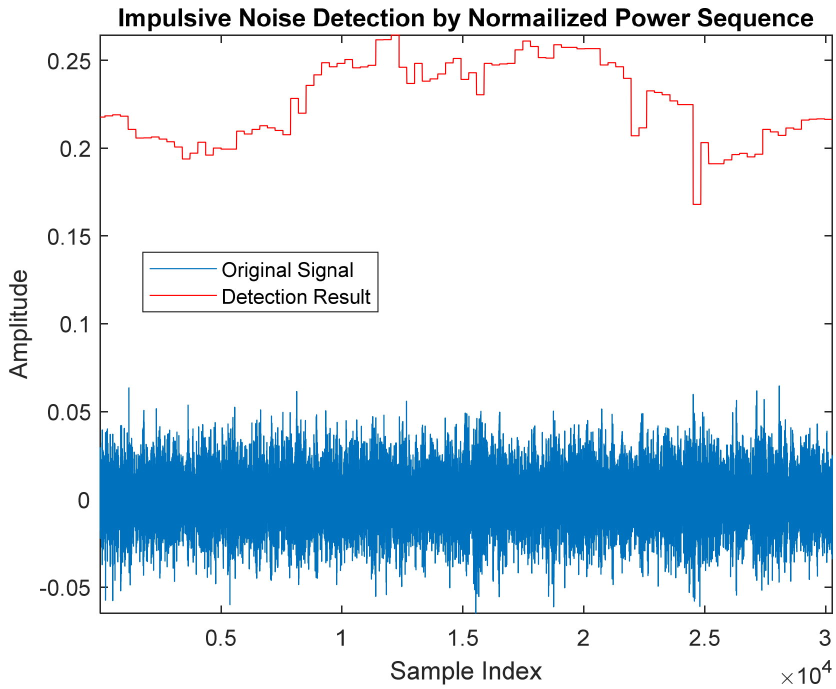

Figure 9. Impulsive noise detection performance when the acoustic environment is free of impulse noise. Since no impulsive characteristics are present, the detection output (red line) shows minimal fluctuations, confirming the algorithm's stability and resistance to false detection under steady-state conditions.

Impulsive noise detection was carried out using a conditional median filter to identify transient disturbances within the signal. As shown in Figure 10, the upper plot represents the original noise signal, the middle plot illustrates the output of the conditional median filter, and the lower plot displays the corresponding detection results. The results demonstrate that the conditional median filter effectively detects impulsive noise components, confirming its suitability for preprocessing in the proposed hybrid ANC system.

Figure 11 presents the detection results obtained using the proposed standard deviation-based rank order mean (SD-ROM) algorithm applied to the impulsive noise signal. The figure illustrates that the SD-ROM algorithm accurately identifies the occurrence and location of impulsive noise events within the signal. The results demonstrate that the proposed algorithm can effectively distinguish impulsive noise components from the background signal, confirming its reliability and robustness for impulsive noise detection in sensitive acoustic environments such as infant incubators.

In this study, a normalized power sequence method is used for impulsive noise detection.

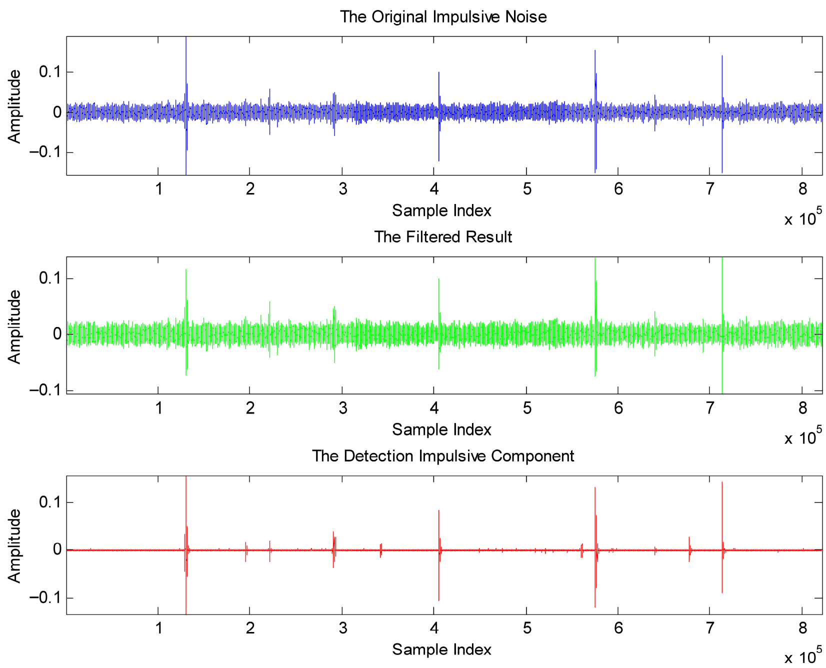

Figure 10. Impulsive noise detection using the conditional median filter. The figure displays three corresponding time-domain plots: (**upper**) the original noise signal; (**middle**) the output of the conditional median filter; and (**lower**) the resulting binary detection signal. These results collectively demonstrate that the conditional median filter can effectively and reliably detect the presence of impulse noise.

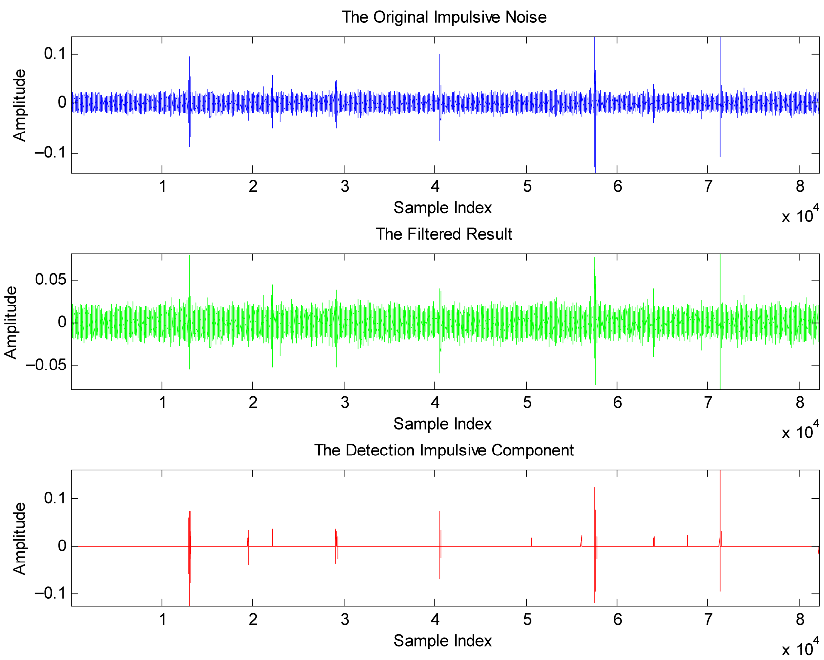

Figure 11. Impulsive noise detection by SD-ROM algorithm. (**upper**) Original noise, (**middle**) output of the SD-ROM method; (**lower**) the detection results. The SD-ROM method can effectively detect the impulse noise.

5.2. Impulsive Noise Control

In the simulation, the initial w(0) is chosen to be 10^{-4} and K(0) is 10^{-8}. v_m (n) is a zero-mean white noise generated by MATLAB. The correlation of it is 1.1×10^{-8}. The performance of FXLMS algorithm is shown in Figures 12 and 13. There is no obvious reduction in the reference noise. The average cancellation for the FXLMS algorithm is

0.063 dB on the left and 0.115 dB on the right. FXLMS is not able to adapt to dramatic changes from impulsive noise.

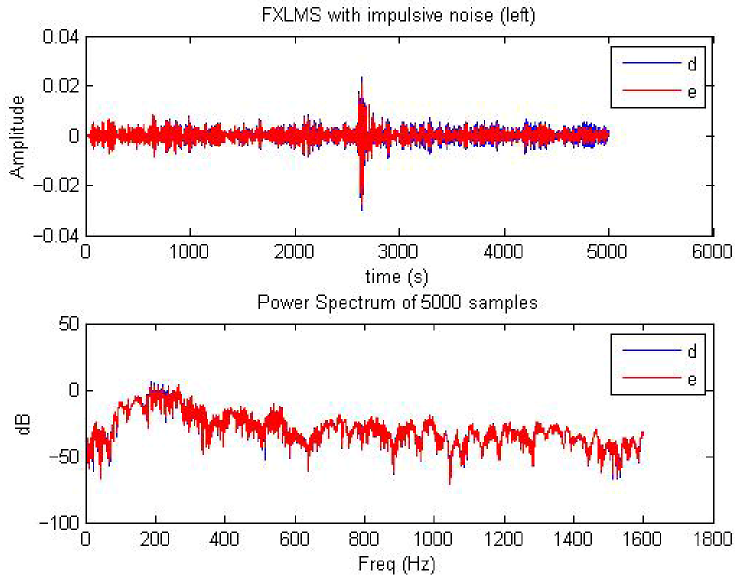

Figure 12. Performance of the Filtered-x Least Mean Squares (FXLMS) algorithm when dealing with impulsive noise (left panel). The overlapping blue (ANC off) and red (ANC on) curves are virtually indistinguishable, resulting in a measured noise cancellation gain of only approximately 0.06. This observation demonstrates the severe limitations of the standard FXLMS algorithm in mitigating impulsive noise.

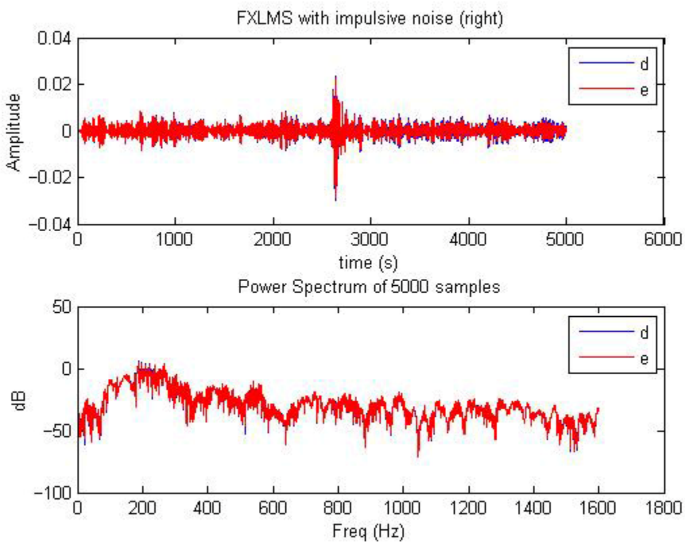

Figure 13. Performance of the Filtered-x Least Mean Squares (FXLMS) algorithm when dealing with impulsive noise (right error sensor). The overlapping blue (ANC off) and red (ANC on) curves are virtually indistinguishable, resulting in a measured noise cancellation gain of only approximately 0.115. This observation demonstrates the severe limitations of the standard FXLMS algorithm in mitigating impulsive noise.

And Figures 14 and 15 show the results of FXNLMS for the same impulsive noise. In FXNLMS [33,34], the length of adaptive filter is 64, the same as in the Kalman filter. And the initial value of the filter is 0. As seen in these figures, the amplitude difference is much smaller than the FxLMS algorithm. As well as the amplitude in the frequency domain, the error signal is close to the original noise. As a result, there is 1.278 dB of cancellation on the left side and 1.181 dB on the right.

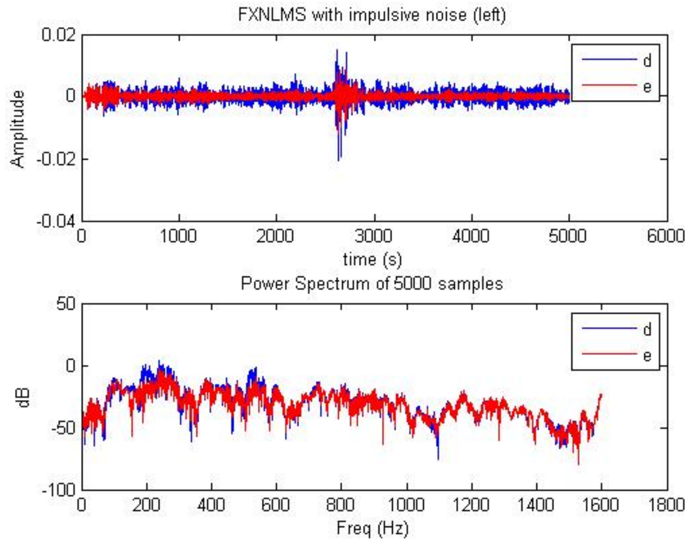

Figure 14. Performance of the Filtered-x Normalized Least Mean Squares (FXNLMS) algorithm when mitigating impulsive noise. The figure presents both (**upper**) time-domain and (**lower**) frequency-domain (magnitude) results. The slight separation between the red (ANC on) and blue (ANC off) curves indicates limited performance improvement, resulting in a measured noise reduction gain of approximately 1.278 dB.

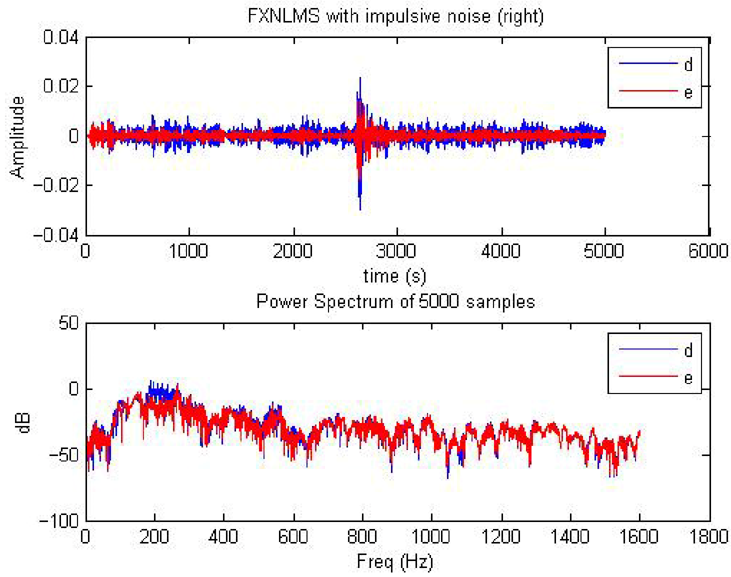

Figure 15. Performance of the Filtered-x Normalized Least Mean Squares (FXNLMS) algorithm when mitigating impulsive noise. The figure presents both (**upper**) time-domain and (**lower**) frequency-domain (magnitude) results. The slight separation between the red (ANC on) and blue (ANC off) curves indicates limited performance improvement, resulting in a measured noise reduction gain of approximately 1.18 dB.

As seen in Figures 16 and 17, the upper one is the results in the time domain. The x-axis represents the sample index and the y-axis is the amplitude of the signal. The blue line represents the original reference noise and the red line is the error signal when the Kalman filter is used to update w(n). The Kalman filter can effectively cancel the impulsive noise as well as the background noise. The bottom one is the spectrum of these two signals. After calculation for average cancellation from 0 Hz to 1600 Hz using 5000 samples, there is 5.1525 dB of cancellation in the left error microphone and 4.2713 dB at right error microphone. Although a 5 dB noise reduction is not sufficient to fully resolve impulsive noise challenges in the NICU. However, it represents a clear improvement over the 0–1 dB reduction produced by the comparison methods, indicating that the proposed approach is more effective in the dynamic, mixed-noise conditions typical of clinical environments.

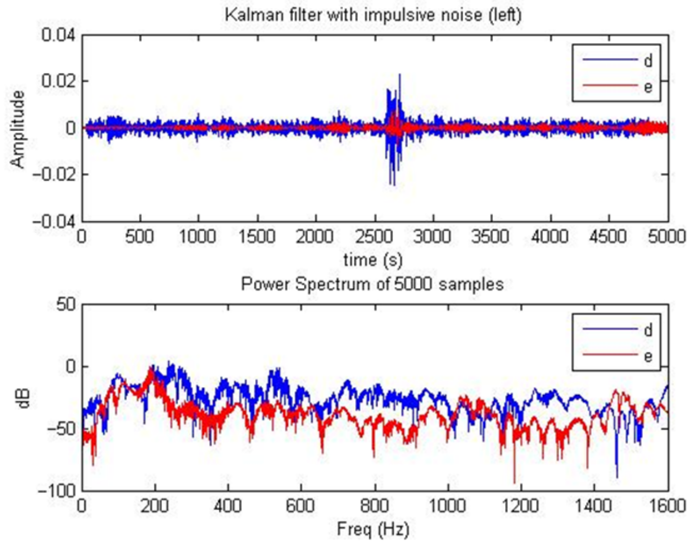

Figure 16. Performance of the adaptive Kalman filter algorithm when mitigating impulsive noise (left error sensor). The figure presents the results in the (**upper**) time-domain and (**lower**) frequency-domain (magnitude). The clear separation between the blue (ANC off) and red (ANC on) curves demonstrates a substantial noise reduction gain of approximately 5.1525 dB, confirming the effectiveness of the proposed approach in managing transient noise.

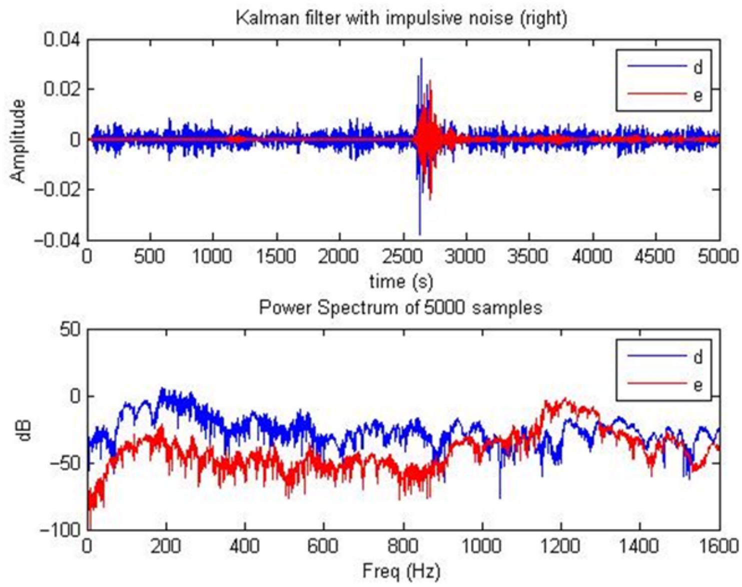

Figure 17. Performance of the adaptive Kalman filter algorithm when mitigating impulsive noise (right error sensor). The figure presents the results in the (**upper**) time-domain and (**lower**) frequency-domain (magnitude). The clear separation between the blue (ANC off) and red (ANC on) curves demonstrates a substantial noise reduction gain of approximately 4.27 dB, confirming the effectiveness of the proposed approach in managing transient noise.

In Figures 18 and 19, the spectra of primary noise, the Kalman filter, FXNLMS, and FXLMS are put together. The blue line represents reference noise, the green line is the spectrum of the Kalman filter, and the red dotted line is the performance of FXNLMS. The cyan line represents the performance of the FXLMS algorithm. As seen in Figures 17 and 18, the differences between the blue lines, dotted red lines, and cyan lines are very small while the green lines can be easily distinguished from the blue lines. And the power of the green ones is dramatically lower than the blue ones. These two figures verify the Kalman filter's tracking capability as a robust replacement of FXNLMS or FXLMS when there is impulsive noise. The Kalman filter can adjust itself to changes caused by sudden impulsive noise and still achieve better cancellation than FXNLMS.

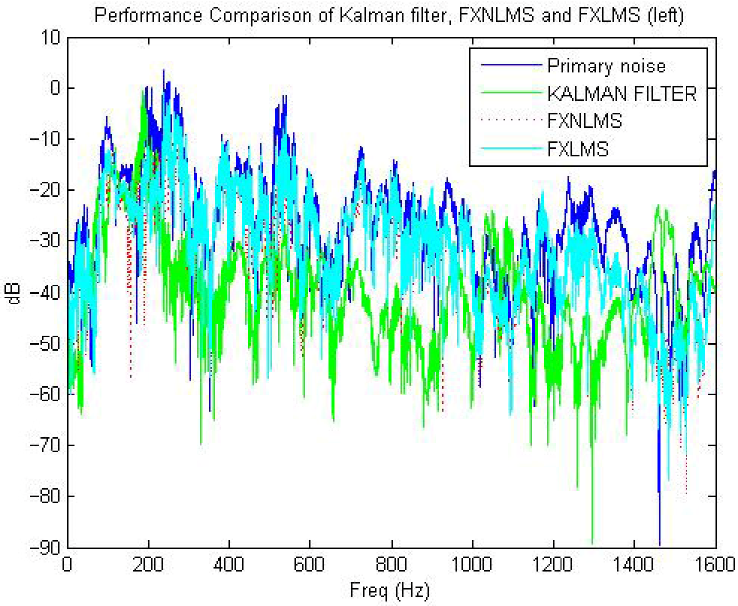

Figure 18. Comparative performance evaluation (left error sensor) of the Kalman filter (KF), Filtered-x Non-linear Least Mean Squares (FXNLMS), and Filtered-x Least Mean Squares (FXLMS) algorithms when suppressing impulsive noise (left panel). The chart clearly demonstrates the superior noise reduction capability of the adaptive Kalman filter 5.15 dB compared to the FXNLMS 1.28 dB and the standard FXLMS 0.06 dB) algorithms, highlighting the necessity of model-based state estimation for mitigating transient acoustic events.

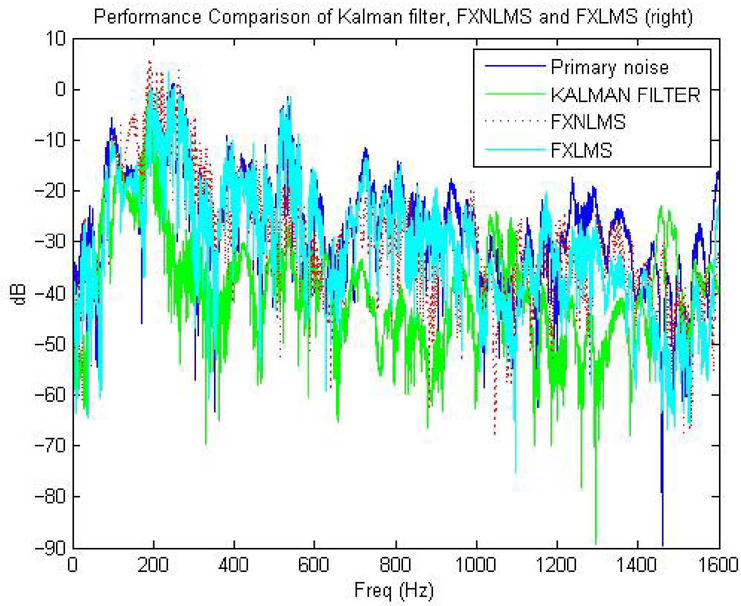

Figure 19. Comparative performance evaluation (right error sensor) of the Kalman Filter (KF), Filtered-x Non-linear Least Mean Squares (FXNLMS), and Filtered-x Least Mean Squares (FXLMS) algorithms when suppressing impulsive noise (left panel). The chart clearly demonstrates the superior noise reduction capability of the adaptive Kalman filter 4.27 dB compared to the FXNLMS 1.18 dB and the standard FXLMS 0.12 dB algorithms, highlighting the necessity of model-based state estimation for mitigating transient acoustic events.

As illustrated in Figure 20, the Kalman filter demonstrates superior performance compared with the FxNLMS and FxLMS algorithms in handling impulsive noise. The blue curve represents the error learning trajectory of the Kalman filter, the green curve corresponds to the FxNLMS algorithm, and the red curve denotes the FxLMS algorithm. The Kalman filter exhibits the fastest convergence rate and the lowest steady-state error power, indicating excellent tracking capability and minimal residual noise. The FxNLMS algorithm achieves moderate performance, converging faster than the conventional FxLMS due to its normalized gradient update, yet it remains less effective than the Kalman filter. In contrast, the FxLMS algorithm shows the slowest adaptation speed and the highest steady-state error, reflecting its limited ability to cope with impulsive or rapidly varying noise. These results confirm that the Kalman filter serves as a superior replacement for FxNLMS in impulsive noise environments, owing to its combined advantages of optimal state estimation, strong tracking ability, and rapid convergence.

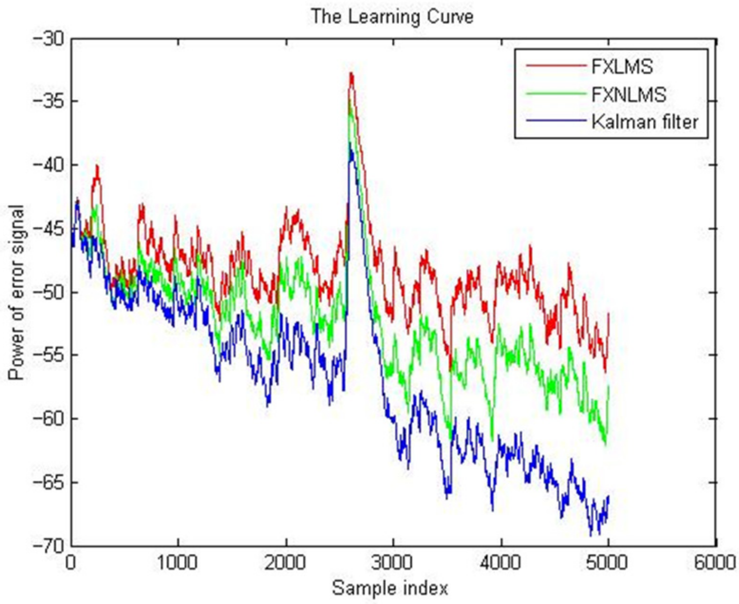

Figure 20. Learning curves of the FxLMS, FxNLMS, and Kalman filter-based ANC algorithms. The Kalman filter achieves the fastest convergence and lowest error power, indicating the smallest residual noise. The FxNLMS performs moderately, while the FxLMS shows the slowest adaptation and highest residual error.

The current results indicate that reducing impulsive noise levels in the NICU can contribute to a calmer acoustic environment, which is associated with improved infant stability and caregiver interaction. This reduction supports the potential for better clinical outcomes by minimizing noise-related stress and enhancing the overall quality of communication within the care setting.

6. Conclusions

A hybrid adaptive active noise control (ANC) system is proposed for application in an infant incubator, representing a highly sensitive acoustic environment. The system integrates impulsive noise detection methods based on normalized power, conditional median filtering, and the standard deviation-based rank order mean (SD-ROM) algorithm to identify and isolate transient disturbances. Once detected, a Kalman filter is activated to suppress impulsive noise, while a conventional Filtered-X Least Mean Squares (FxLMS) algorithm is employed to attenuate continuous background noise. By combining the robustness of Kalman filtering with the adaptability of the FxLMS algorithm, the proposed hybrid system achieves effective noise reduction under both transient and steady-state conditions. The results demonstrate that this approach enhances acoustic comfort and safety within the incubator environment, highlighting its potential for broader applications in noise-sensitive healthcare and biomedical systems.

The limitations of the current implementation center on performance degradation that may occur under specific, challenging conditions. For example, in the presence of highly transient impulsive noise events, there may be rapid non-linear variations within the secondary path or in frequency regions where the adaptive filter exhibits limited response capability. Future efforts will primarily focus on validating the methodology through real-time hardware implementation and system integration to support practical deployment in a clinical environment. Additional work is planned to incorporate advanced statistical performance measures, rigorously evaluate computational complexity for embedded system use, and expand testing with significantly larger NICU acoustic datasets to further validate the system's long-term robustness and generalizability.

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Conflicts of Interest: The authors declare no conflicts of interest.

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